

Howard Lin

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Education

University of California, San Diego

La Jolla, California

B.S. in Computer Science, expected June 2026. GPA: 3.85

Sep. 2024 - Current

- Relevant Coursework: C++ Programming, Data Structures/Algorithms, Calculus I-III/Multi-variable Calculus, Linear Algebra, Software Construction, Discrete Structures, Machine Learning, Computer Graphics, Operating Systems

Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript

Frameworks/Libraries: React, Flask, Numpy, Pandas, Sklearn

Tools: Git, Github, VS Code, Supabase, Render

Communication: English, Mandarin Chinese

Experience

Software Engineer Intern

City of Industry, California

MSI Computer Corp. (MSI USA)

June 2025 – September 2025

- Developed a full-stack Flask application with Python web-scraping, LLM-powered summarization (Groq API), Supabase SQL storage, and a TypeScript/HTML/CSS frontend deployed on Render
- Built an automated PDF to CSV conversion tool that extracts structured data and streamlines workflows into Google Sheets/Excel
- Partnered with designers to modernize the Student Program landing page, improving UI clarity and user experience

Coding/Robotics Instructor

San Marino, California

Magikid Robotics Lab

July 2023 – January 2024

- Lead summer robotics classes for elementary/middle school students
- Organized and prepared class material for lessons, and managed classes of about 3-6 students
- Taught VexGo, Lego WeDo 2.0, Python, Scratch

Math Tutor

Remote

MyTutor Learning

April 2023 – June 2023

- Assisted students with homework and conceptual questions
- Prepared custom teaching material to engage with individual students
- Analyzed each student's weaknesses to properly and effectively provide support and assistance

Personal/ Team Projects

Heart Disease Predictor | Python, Pandas, NumPy, Scikit-learn, Seaborn

- Led a team of 5 to complete a quarter long machine learning project
- Utilized pandas to read and clean a large CSV dataset for analysis and modeling
- Utilized seaborn to create visual plots and graphs for exploratory data analysis
- Trained a logistic regression model on the data and achieved a precision of 96% for identifying negative heart disease cases and a recall of 70% for positive heart disease cases

Music/Movie recommender | C++, Python, Valgrind, Gtest

- A quarter long project developed in a team of 4
- Utilized different development methods such as Gtest for unit testing, Valgrind for memory leaks, and Python and C++ for back-end development
- Able to take in user specific criteria, filters, and preferences, pass that data into IMDb and Spotify's API, and return a movie or song recommendation
- The project was a collaborative effort and refined communicative as well as team working skills

Pixel Sketch | HTML, CSS, JavaScript

- Customizable drawing web-application with different colors, eraser option, and grid size adjuster
- Incorporated JavaScript animations for front end design, as well as HTML/CSS for structure
- Deployed with GitHub pages